

Photoevaporation in Hot Jupiter Atmospheres

Traced by $\text{Ly}\alpha$, $\text{H}\alpha$, and $\text{He I } \lambda 10830$ Triplet Lines

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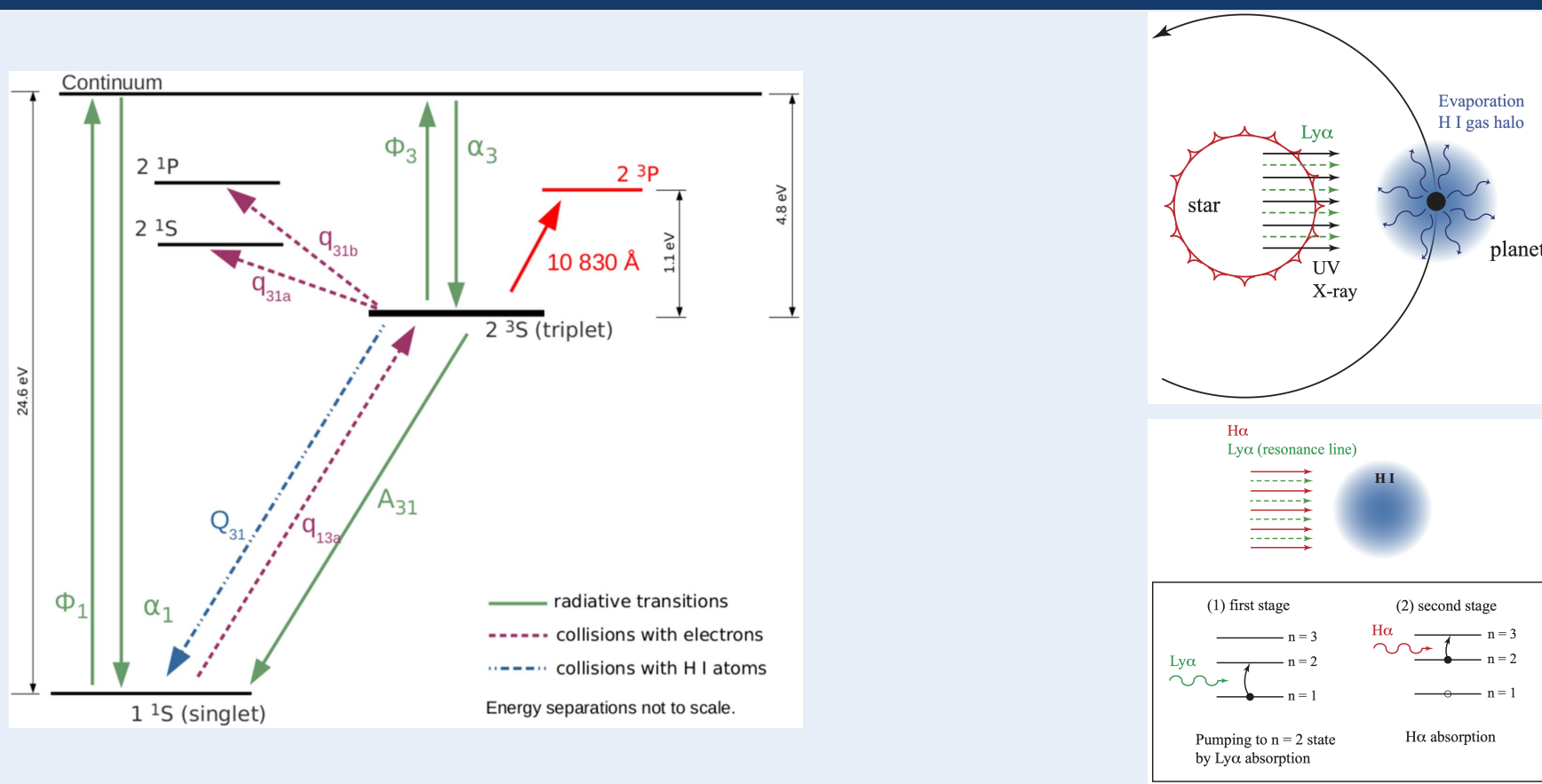
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1. Motivation: three lines, three layers

- Close-in giants lose mass through **XUV-driven hydrodynamic escape** (photoevaporation). Resonance/metastable lines probe the escaping gas at *complementary* layers:
 - $\text{Ly}\alpha$** (H $1s$) – bulk neutral H (resonant; core lost to ISM).
 - $\text{H}\alpha$** (H $n=2$) – $\text{Ly}\alpha$ -pumped excited H; optical.
 - $\text{He I } \lambda 10830$** (He 2^3S metastable) – near-IR.
- Modeling the lines *jointly* constrains the H/He composition, the mass-loss rate, and the (usually unobserved) stellar $\text{Ly}\alpha$ flux.
- We model the inflated hot Jupiter **WASP-52b** (K2V host) with the **EXHALE** escape code coupled to the **LaRT** $\text{Ly}\alpha$ Monte-Carlo code.

2. Line formation: He 2^3S & $\text{H}\alpha$ $\text{Ly}\alpha$ -pumping



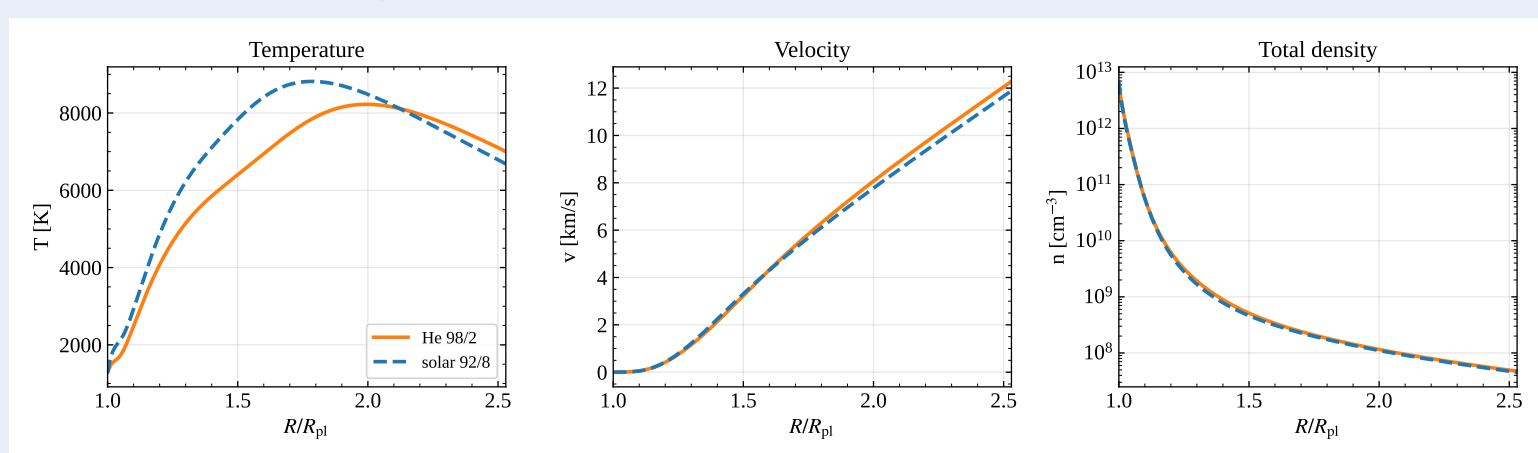
He 10830 (left): metastable He 2^3S ($2^3S \rightarrow 2^3P$), fed by recombination+cascade, drained by FUV photoionization. **$\text{H}\alpha$** (right; $n=2 \rightarrow 3$) needs an enhanced $n=2$ set mainly by **$\text{Ly}\alpha$ pumping** of $1s \rightarrow 2p$.

3. EXHALE: code & algorithms

- 1-D multi-species radiation-hydrodynamics** (extended ATES, Caldiroli+2021): mass/momentum/energy Euler equations $\rightarrow$ steady state, with the full **Roche** potential (tidal + centrifugal; L_1 -truncated or extended).
- Ionization equilibrium** of H, He + metals (C,N,O,Mg,Si,Ca,Na,K,S,Fe) via Newton/MINPACK; photoionization, collisional ionization, (Badnell) radiative+dielectronic recombination, charge exchange.
- Heating** XUV photoionization; **cooling** closed-form CHIANTI metal lines + $\text{Ly}\alpha$ + recombination + bremsstrahlung.
- He 2^3S metastable network** $\rightarrow$ He 10830; **H $n=2 \rightarrow 3$** $\text{H}\alpha$ (Christie+2013 NLTE + $\text{Ly}\alpha$ pumping).
- Steady solver**: PLM $\rightarrow$ WENO3 march + Newton/JFNK (flux convergence); **transmission** via impact-parameter Voigt LOS + disk average (TPM).
- $\text{Ly}\alpha$ RT**: coupled to **LaRT** 3-D Monte Carlo (Seon & Kim 2020) $\rightarrow$ cylindrical $P_\alpha(\rho, z) \rightarrow$ 2-D $n_{2p} \rightarrow \text{H}\alpha$.

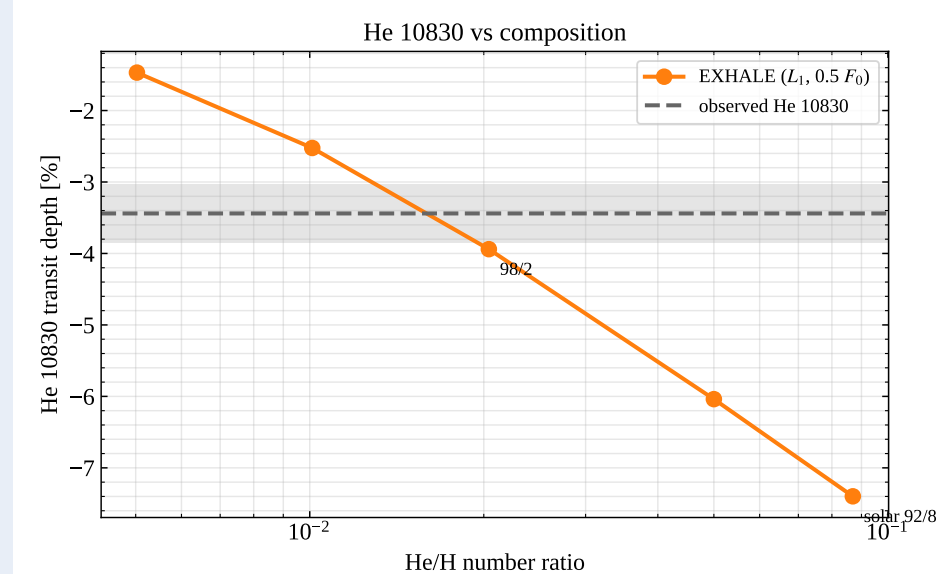
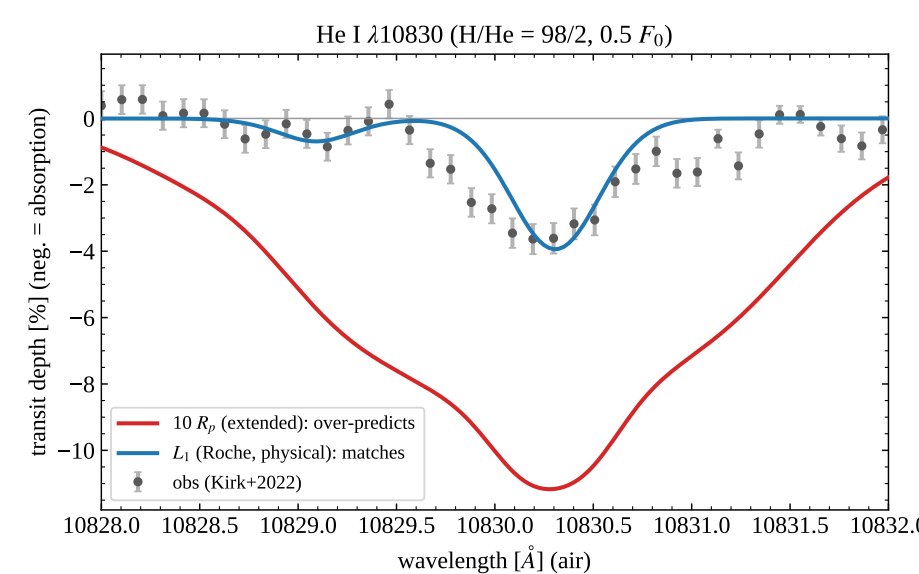
4. WASP-52b model setup

- Planet/star (Yan+2022): $M_\star = 0.87 M_\odot$, $R_\star = 0.79 R_\odot$ (K2V); $M_p = 0.46 M_J$, $R_p = 1.27 R_J$, $T_{\text{eq}} = 1304 \text{ K}$, $a = 0.0272 \text{ au}$.
- Stellar SED: ϵ Eri (MUSCLES) scaled to the orbit; $F_{\text{XUV}} = 3.4 \times 10^4 \text{ erg cm}^{-2} \text{ s}^{-1}$, hardness $\beta_m \equiv F(1-100 \text{ \AA})/F(1-912 \text{ \AA}) = 0.218$.
- Lower boundary pressure-anchored at $1 \mu\text{bar}$; full Roche potential (L_1 at $\sim 2.53 R_p$). Converged wind: T peaks $\sim 10^4 \text{ K}$ near $2 R_p$, transonic, density falling ~ 7 dex over $1-2.5 R_p$.
- Validation**: the steady-state mass-loss rate reproduces Yan+2022 (Table 4) to within $\sim 10\%$ at every XUV flux.



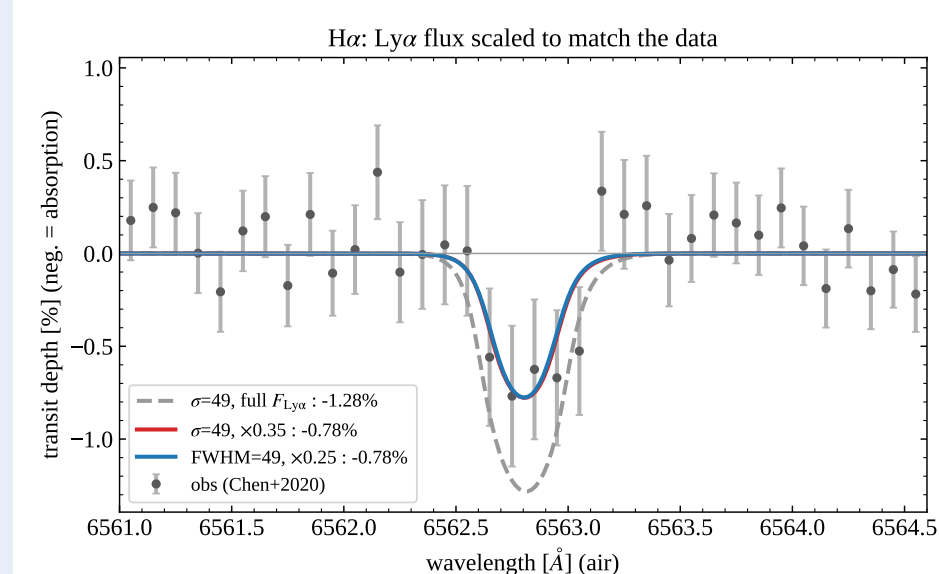
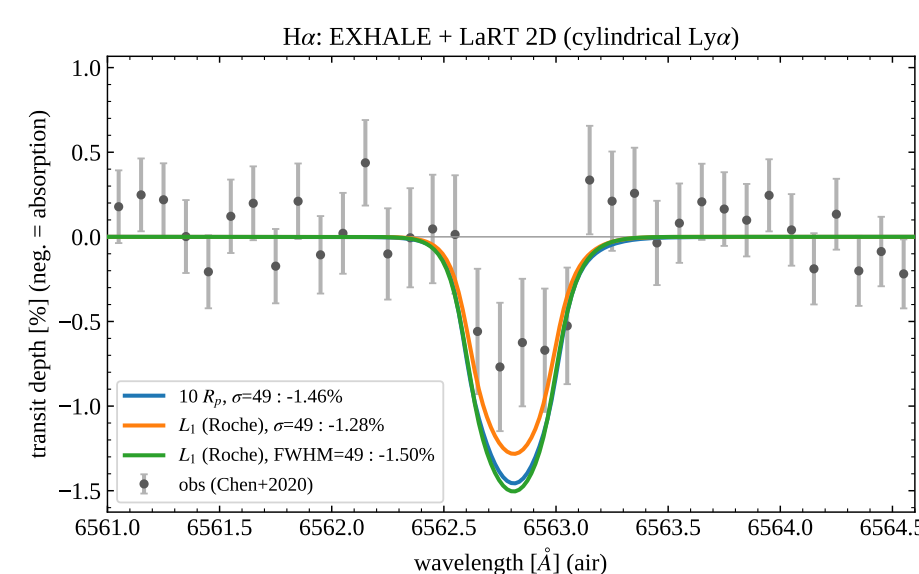
Best-fit wind (**H/He 98/2**, L_1 domain; solar shown for reference).

5. He I $\lambda 10830$: domain & composition



- Domain matters**: as in Yan+2022, extending to $10 R_p$ over-predicts the depth by $3-10\times$ (72% of the column from $r > L_1$, where 1-D spherical over-counts the L_1 funnel); the physical L_1 (**Roche**) domain matches.
- Composition**: solar (92/8) is too strong; matching the observed $\sim 3.4\%$ requires a **He-depleted** atmosphere, **H/He $\approx 98/2$** (He/H ≈ 0.017).

6. $\text{H}\alpha$ via proper $\text{Ly}\alpha$ radiative transfer (LaRT)



The **$\text{Ly}\alpha$ scattering (pump) rate** P_α [$\text{s}^{-1} \text{ atom}^{-1}$] is built from the LaRT per-luminosity rate $\mathcal{P}_{\text{LaRT}}(\rho, z)$:

$$P_\alpha \equiv B_{1s,2p} \bar{J}_\alpha = \mathcal{P}_{\text{LaRT}} L_{\text{Ly}\alpha} \frac{\Omega_\star}{4\pi}, \quad L_{\text{Ly}\alpha} = \frac{F_{\text{Ly}\alpha} 4\pi a^2}{h\nu_\alpha}.$$

The excited $n=2p$ population follows the Christie+2013 $2s/2p$ equilibrium,

$$n_{2p} \simeq \frac{(P_\alpha + C_{1s2p}n_e)n_{1s} + \alpha_{2p}n_e^2}{A_{2p1s} + (C_{2p1s} + C_{2p2s})n_e + \Gamma_{2p}} \xrightarrow{\text{pump-dom.}} n_{1s} \frac{P_\alpha}{A_{2p1s}},$$

($\bar{J}_\alpha = P_\alpha/B_{1s,2p}$; $\Omega_\star/4\pi$ = stellar solid-angle dilution; Γ_{2p} = Balmer-continuum photoionization.)

- Stellar illumination makes P_α (hence n_{2p}) **cylindrically symmetric** $\Rightarrow$ a 2-D $n_{2p}(\rho, z)$ integrator; incident $\text{Ly}\alpha$ double-Gaussian ($\pm 74 \text{ km/s}$, width 49 km/s).
- $\text{H}\alpha$ is **pump-dominated**, domain-*insensitive*, depth $\propto F_{\text{Ly}\alpha}$; matched at $F_{\text{Ly}\alpha} \approx 0.3-0.5\times$ the EP Eri proxy $\Rightarrow$ WASP-52's $\text{Ly}\alpha$ is $\sim 1/3-1/4$ of the proxy.

7. Parameters that reproduce the observations

Observable	Reproduced when
Mass loss $\dot{M}$	matches Yan+2022 within $\sim 10\%$ (<i>no tuning</i>)
He 10830 ($\sim 3.4\%$)	L_1 (Roche) domain + He-poor H/He $\approx 98/2$ (He/H ≈ 0.017) + $0.5 F_0$
$\text{H}\alpha$ ($\sim 0.8\%$)	LaRT $\text{Ly}\alpha$ RT + $F_{\text{Ly}\alpha} \approx 0.3-0.5\times$ EP Eri proxy
XUV hardness β_m	0.218

8. Conclusions

- EXHALE+LaRT reproduce WASP-52b's He 10830 *and* $\text{H}\alpha$ with one self-consistent escape model; He 10830 demands the **Roche (L_1) domain** and a **He-depleted** atmosphere (H/He $\approx 98/2$).
- $\text{H}\alpha$, via proper $\text{Ly}\alpha$ RT, is a **diagnostic of the stellar $\text{Ly}\alpha$ flux** ($\approx 1/3-1/4$ of the EP Eri proxy). Jointly modeling the three lines breaks degeneracies among composition, $\dot{M}$, and stellar $\text{Ly}\alpha$.
- Outlook**: we will extend this EXHALE+LaRT framework to a **systematic analysis of other exoplanet atmosphere observations** (He I 10830 / $\text{H}\alpha$ / $\text{Ly}\alpha$), building a uniform grid of escape models.

Observations: He I 10830 – Kirk et al. 2022 (AJ 164, 24; Keck/NIRSPEC); $\text{H}\alpha$ – Chen et al. 2020 (A&A 635, A171; ESPRESSO/VLT).

Model/theory: Yan, Seon, Guo, Chen & Li 2022 (ApJ 936, 177); Caldiroli+2021 (A&A 655, A30); Yang & Guo 2018 (RAA 18, 077);

Christie+2013 (ApJ 772, 144); Huang+2017 (ApJ 851, 150); Seon & Kim 2020 (ApJS 250, 9).